

Amelia Eckard

Charlotte, NC | ameliaeckard.com | github.com/ameliaeckard | linkedin.com/in/ameliaeckard

Education

University of North Carolina at Charlotte <i>M.S. in Artificial Intelligence / Early Entry Program</i>	Charlotte, NC <i>Expected May 2028</i>
University of North Carolina at Charlotte <i>B.S. in Computer Science / Concentration: AI, Robotics, & Gaming</i>	Charlotte, NC <i>Expected May 2027</i>

Experience

Undergraduate Researcher <i>UNC Charlotte, Dr. Todd Dobbs</i>	Charlotte, NC <i>August 2025 – May 2026</i>
- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues. - Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.	

Research Intern <i>Stevens Institute of Technology, Dr. Jina Huh-Yoo</i>	Remote <i>June 2026 – Present</i>
Engineered <i>rq-system</i>/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers.	

Projects

CNN Scene Classification <i>Python, PyTorch, torchvision, CUDA</i>	2026
Trained and evaluated convolutional neural networks for 16-class scene recognition on 2,400 images.	
Vision Lab: ASCII Reconstruction <i>JavaScript, HTML/CSS, Image Processing</i>	2026 – Present
- Developed an image-to-ASCII reconstruction experiment for a web-based computer-vision lab connected to ameliaeckard.com. - Designed reconstruction logic around luminance mapping, local contrast, detail preservation, and edge-aware character selection to retain recognizable image structure.	

Technical Skills

Languages	Swift, Python, C++, JavaScript
Spatial/AI	ARKit, RealityKit, Core ML, OpenCV, ObjectTrackingProvider, HRTF spatial audio
Tools	Git, CUDA, NumPy, PyTorch, HTML/CSS

Certifications & Involvement

UR2PhD Training Course, Computing Research Association	December 2025
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